Predicting Wind Turbine Failures Before They Happen
A Cost-Optimised Classification Approach

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Business Recommendation

Deploy Model 5

Bundle StandardScale with the Model

Monitor sensors V18, V21, V15, v7

Executive Summary

Business Interpretation of the Results

248 Failures caught (True Positives)

Turbines flagged before failure and repaired on schedule. Repair cost incurred – significantly lower than replacement. This is the optimal outcome

~8 false alarms (False Positives)

Healthy turbines flagged for inspection. Tech dispatched, no fault found. Inspection cost only – The lowest cost outcome. At only 8 false alarms on 5,000 predictions, this is exceptionally low and operationally manageable

Precision = TP / (TP + FP)
0.97 = 248 / (248 + FP)

248 + FP = 248 / 0.97
248 + FP = 255.67
FP = 255.67 - 248
FP ≈ 7.6 → ~8

~34 missed failures(False Negatives) -- primary risk

Turbines that will fail without a maintenance flag. Full replacement cost incurred. These are the primary remaining business risk and the target for future model improvement via SMOTE. In preliminary tests it achieved better rates.

TP = 0.88 × 282 = 248.16 → ~248 failures correctly caught
FN = 282 - 248 = 34 → ~34 failures missed

Business Problem Overview and Solution Approach

Problem Definition

ReneWind collects data from 40 sensors fitted across wind turbine generators to predict mechanical failure before it occurs. ReneWind needs to find the best way to implement predictive maintenance for turbine generators based on the high-density sensor network data provided

Sensor Array:

40 telemetry features (V1 - V40) monitoring turbine health.

Dataset Scale:

25,000 total observations (20k Train / 5k Test).

The "Needle in the Haystack" Challenge:

Class Imbalance:

17:1 ratio of Normal operation to Failure.

Minority Class:

Only 5.55% of the data represents actual failure events.

Business Problem Overview and Solution Approach

Cost-Benefit Approach

In wind energy, not all errors are equal. Our model logic is dictated by the cost of being wrong.
The cost structure is asymmetric. A missed failure leads to unplanned breakdown and full generator replacement — the most expensive outcome. A correct prediction enables a scheduled repair at moderate cost. A false alarm triggers only a cheap inspection.
This hierarchy makes **Recall the primary metric** — catching failures matters more than avoiding false alarms.

Scenario	Event	Outcome	Cost Impact
False Negative	**Missed Failure**	**Unplanned Breakdown**	🔴 **Critical** (Full Replacement)
True Positive	**Correct Prediction**	**Scheduled Repair**	🟡 **Moderate** (Maintenance)
False Positive	**False Alarm**	**Unnecessary Inspection**	🟢 **Low** (Technician Visit)

Business Problem Overview and Solution Approach

Please mention the solution approach/methodology

The objective is to build various classification models, tune them, and select the best one to identify failures, so the generators can be repaired before failing or breaking, reducing overall maintenance costs. Considering the cost structure is asymmetric and drives every modelling decision, the nature of predictions made by the classification model will translate as follows:

Missed failure (False Negative)

Results in unplanned generator breakdown and full replacement — the highest cost outcome

Correct prediction (True Positive)

Triggers a scheduled repair — moderate cost, significantly cheaper than replacement

False alarm (False Positive)

Rriggers an unnecessary inspection — the lowest cost outcome

EDA Results

The training dataset

Dataset Dimensions:

20,000 training records | 5,000 test records.

Feature Profile:

40 numerical sensor inputs (float64) and 1 binary integer target.

Class Definition:

0 (Normal Operation) vs. 1 (Equipment Failure).

Data Integrity:

38 of 40 features were 100% complete.

Missing Value Strategy:

Features V1 and V2 contained missing values(approx 0.09). These were resolved via Median Imputation to maintain the integrity of the feature distributions while being robust to potential outliers.

Metadata Quality:

For security reasons, the feature names were scrambled, creating meaningless names. It diffuses the quality of the analysis due to a lack of better context

Link to Appendix slide on data background check

Data Preprocessing

Duplicate value check

No Duplicate value found in Train and Test.

Missing value treatment

The training data features V1 and V2 contained missing values(approximately 0.09). These were resolved via Median Imputation to maintain the integrity of the feature distributions while being robust to potential outliers. We applied the same procedure to missing V1 and V2 features in the test data.

Data Preprocessing

Outlier check (treatment if needed)

The outlier rates are low but not zero. V34 is the highest in the dataset. The spikes are important; we kept the outliers because these are the way for the turbine to notify that it is about to break. V18 and V15 are in the top 5 Predictors. They are the most correlated with failure. If we removed then we would get rid of the most informative signal in the dataset.

Relu also handles outliers well, another reason to keep them.

Feature	Outlier %
V34	4.02%
V18	3.66%
V15	2.57%
V33	1.92%
V29	1.68%
V35	1.58%
V24	1.54%
V13	1.52%
V17	1.48%
V7	1.46%

Data Preprocessing

Feature engineering

Two transformations were applied to V1 and V2 using median imputation. It replaced the 18 values in the training dataset and 5 and 6 values in the test dataset.

StandardScaler normalization transformed all 40 features, changing their basic scale. It helped with gradient descent. This was not in the low-code notebook. I applied it to enhance Recall using SMOTE, which I will briefly comment on at the end.

Feature engineering was intentionally minimal.

We considered applying dimensionality reduction because of the 40 features. It was not applied because NN handles these features well.

Feature vs Feature correlation is <0.3, mostly low. There is low redundancy and high density.

Feature vs Target correlation is small; It requires a deep learning model to find patterns

Data Preprocessing

Data preparation for modeling

The target variable is severely imbalanced — 18,890 observations (94.45%) belong to class 0, and only 1,110 (5.55%) to class 1, resulting in a 17:1 ratio.

Balance class weights were computed using sklearn compute_class_weight at the rate [0.52, 9.009] to invert the imbalance contribution mentioned above.

Model Evaluation Criterion

Because a missed failure costs significantly more than a false alarm, we cannot rely on standard "Accuracy." We use a specific set of tools to measure performance.

Primary metric Recall

To measure the proportion of actual failures correctly identified by the model. It is the primary business metric because the cost of a turbine is much higher than maintenance costs.

Recall = TP / (TP + FN).

At threshold 0.3: 248 of 282 failures caught → Recall = 0.88

Secondary metric Precision

To measure the proportion of alerts that are genuine failures. It cannot be ignored, as it can cause up to 4718 unnecessary maintenance trips. At that point, the overall cost may surpass the cost of a turbine.

Precision = TP / (TP + FP)

At threshold 0.3: only ~8 false alarms out of ~256 alerts → Precision = 0.97

Model Evaluation Criterion

Because a missed failure costs significantly more than a false alarm, we cannot rely on standard "Accuracy." We use a specific set of tools to measure performance.

Support metric F1

IT is the harmonic mean of precision and recall. F1 penalizes extreme imbalance between the two. It was not used as a primary metric because it assumes equal cost for false positives and false negatives. For ReneWind false negative is extremely costly for it would mean to replace a turbine.

Reference metric Accuracy

It was not used for model selection due to class imbalance.

Threshold Tuning

Used to maximize Recall, given we want to consider even the borderline fail possibility.

The default threshold of 0.5 was reduced to 0.3 across all final evaluations. This adjustment reflects the cost asymmetry directly — at 0.3, the model acts on lower-confidence failure signals, catching borderline cases that would otherwise be missed. The table below shows the effect of the threshold on the key metrics:

Threshold	Recall	Precision	F1	False alarms	Missed failures
0.5	0.86	0.96	0.91	~10	~39
0.3	0.88	0.97	0.92	~8	~34

Model Evaluation Criterion

Best Model on Test Results

Metric	Role	Formula	Final value
Recall	Primary — minimise missed failures	TP / (TP + FN)	0.88
Precision	Constraint — limit false alarms	TP / (TP + FP)	0.97
F1 score	Balance check	2×P×R / (P+R)	0.92
Accuracy	Reference only — not used for selection	(TP+TN) / Total	0.98

Model Performance Summary

	Model 0	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Accuracy	0.989250	0.990750	0.989500	0.989000	0.991000	0.992250	0.988250
Recall	0.909513	0.929386	0.924485	0.943299	0.933758	0.936540	0.945022
Precision	0.986649	0.980696	0.972933	0.950956	0.978684	0.988704	0.943144
F1 Score	0.944343	0.953407	0.947230	0.947090	0.954971	0.960963	0.944080

The average time to run was 30s, with SGD models slightly faster than ADAM at 26s.

Model 5 is the best all-around performance. It leads to Accuracy, Precision, and F1. With Recall At 0.9365, it is just below the top 2. The combination makes it the most reliable model.

Model 6 leads in recall but has the lowest precision of all; it generates more false alarms.

Model 3 offers an interesting middle ground; it is more aggressive for failure detection with slightly more false alarms.

Link to Appendix slide on model assumptions

Best Model - Test Performance

Architecture of the best model

INPUT	HIDDEN LAYER 1	REGULARIZATION	HIDDEN LAYER 2	HIDDEN LAYER 3	OUTPUT
40	128	0.5	64	32	1
Sensor features	ReLU	dropout	ReLU	ReLU	Sigmoid

Comments on the best performance model on the test set

It was evaluated on the held-out test set of 5,000 observations at a classification threshold of 0.3, which was chosen to capture all possible

The model performed well, capturing 88% of failures, achieving 97% real alert rate, and achieving the best-balanced F1 score of 0.92. It achieved approximately 8 false alarms out of 5000 predictions.

Metric	Class 0 (No failure)	Class 1 (Failure)	Macro avg
Precision	0.99	0.97	0.98
Recall	1.00	0.88	0.94
F1 score	1.00	0.92	0.96
Support	4,718	282	5,000

k to Appendix slide on model assumptions

APPENDIX

Slide Header

- Please add any other pointers (if needed)

	Model 0	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Accuracy	0.98750	0.98950	0.99000	0.98600	0.98700	0.99300	0.98300
Recall	0.90011	0.91389	0.92687	0.93747	0.94224	0.94118	0.94012
Precision	0.97793	0.98438	0.97559	0.93018	0.93487	0.99153	0.90590
F1 Score	0.93513	0.94601	0.94974	0.93379	0.93852	0.96482	0.92221

Happy Learning !